

PROBLEMS ON ELECTRIC AND MAGNETIC MATERIALS

1.- At what temperature will a silver wire exhibit the same resistivity as an aluminium wire at 45 °C. The resistivities of silver and aluminium at 0 °C are $1.47 \cdot 10^{-6}$ and $2.70 \cdot 10^{-6} \Omega \cdot \text{cm}$ respectively and their resistivity coefficients, 0.0038 and $0.0039 \text{ } ^\circ\text{C}^{-1}$, respectively.

(Solution: 305 °C)

2. Reason what effect the following strategies would have on the conductivity and tensile strength of an aluminum alloy wire:

- a) Alloying with a more conductive metal as copper.
- b) Decrease on its working temperature.
- c) Thermal treatment that increases the grain size of the alloy.
- d) Treatment that causes intermetallic precipitation.
- e) Metal drawing (forming) that involves a high % of cold work.

Indicate which would be the best strategy to reduce the weight of the wire while maintaining its tensile strength and conductivity.

(Solution: a) conductivity: decrease; strength: increase; b) conductivity: increase; strength: more increase; c) conductivity: increase; strength: decrease; d) conductivity: decrease; strength: increase; e) conductivity: decrease; strength: increase)

3. Compare the effect of following factors on the conductivity of an ionic conductor and a metallic conductor

	σ - Ionic ceramic	σ - metal
Vacant positions in the net		
Cations in interstitial positions		
Anions in interstitial positions		
Increase of temperature		
Disorder of the net		
Doping with substances with other valences		

4. Justify the different effects that an increase on the exposure temperature have on a metal and on an intrinsic semiconductor

5.- Indicate which of the following semiconductors are extrinsic n-type and extrinsic p-type and whether the introduced impurities act as donors or acceptors

- a) Silicon doped with antimony in substitutional solid solution
- b) Silicon doped with arsenic in substitutional solid solution
- c) Silicon doped with gallium in substitutional solid solution

(Solution: a) n-type extrinsic; b)) n-type extrinsic; c) p-type extrinsic)

6.- P is added to high purity silicon in order to introduce a concentration of 10^{+23} m^{-3} of charge carriers at room temperature.

- a) What type of semiconductor is this material?
- b) What is the conductivity of this semiconductor at room temperature?

Assume that the mobility of the e^- in the material is $0.14 \text{ m}^2/(\text{V}\cdot\text{s})$

(Solution: b) $2243 \Omega^{-1}\text{m}^{-1}$)

7.- The electrical conductivity of intrinsic germanium at 25°C is $2.22 \Omega^{-1}\text{m}^{-1}$. What will its conductivity be at 75°C ? The band gap of germanium is 0.67 eV and $k_B = 8.62 \cdot 10^{-5} \text{ eV/K}$

(Solution: $14.4 \Omega^{-1}\text{m}^{-1}$)

8.- The electrical conductivity of intrinsic Si at room temperature is $4 \cdot 10^{-4} \Omega^{-1}\text{m}^{-1}$, and the mobilities of electrons and holes are 0.14 y $0.048 \text{ m}^2/(\text{V s})$ respectively. Calculate the concentration of holes and electrons.

(Solution: $1.32 \cdot 10^{16} \text{ m}^{-3}$)

9. Order the following semiconductors according to their conductivity (assume that the materials are always at room temperature, except when the opposite is specified).

- a) Pure Germanium at 0°C ; pure Germanium at 25°C ; pure Germanium at 50°C .
- b) Phosphor doped- silicon; pure silicon.
- c) Germanium doped with 100 ppb of nitrogen; germanium doped with 1ppm of nitrogen.

10.- Order the following semiconductors according to their conductivity (assume that the materials are always at the same temperature).

- a) Gallium arsenide; aluminium phosphide; Indium antimonide.

- b) Cadmium telluride; Indium antimonide.
- c) Germanium with 1ppm of P; Germanium with 1 ppm of Sb.
- d) Germanium with 1ppm of P; Silicon with 1 ppm of P.
- e) Silicon with 10 ppm of B; Silicon with 10 ppm of In.
- f) Germanium with 5ppm of As; Germanium with 5 ppm of Sb.

11- Draw the hysteresis curve of a soft magnetic material and a hard magnetic material and indicated in both cases the most significant features. What are the main differences between these two materials?

12.- Specify in the following table how the increase in the following microstructural parameters affect the listed properties of a ferromagnetic material

	Grain size	Impurities	Porosity
Hysteresis losses			
Conductivity			
Permeability			
Coercitivity			
Saturation Magnetization			
Curie Temperature			